



Responses of plants to polybrominated diphenyl ethers (PBDEs) induced phytotoxicity: A hierarchical meta-analysis

Yuan Sun¹, Cuiting Wang¹, Xuan Xu, Honghua Ruan^{*}

College of Biology and the Environment, Nanjing Forestry University, Nanjing, 210037, PR China

HIGHLIGHTS

- PBDEs treatments initiated significant toxicity to plants.
- Lower congener PBDEs were more toxic to plants than higher congener PBDEs.
- Higher plants were more resistant to PBDEs induced phytotoxicity than lower plants.
- GST activities showed a positive trend with increasing PBDEs concentration.

ARTICLE INFO

Article history:

Received 24 June 2019

Received in revised form

18 August 2019

Accepted 13 September 2019

Available online 14 September 2019

Handling Editor: A. Gies

Keywords:

PBDEs

Toxicity

Meta-analysis

Glutathione S-transferase

Ecotoxicology

ABSTRACT

Biologists have extensively investigated the toxicity of polybrominated diphenyl ethers (PBDEs) on plants in ecosystems, where experiments revealed that PBDEs can promote, inhibit, or have no significant effects on the physiological and biochemical functionality of plants. These studies have stimulated many theoretical works that aimed to elucidate the differences in the toxicity of PBDEs on various plants. However, there has been no quantitative attempt to reconcile theory with the results of empirical experiments. To close this gap between theory and experiments, we conducted a hierarchical meta-analysis to examine the toxicity of PBDEs on plants and confirmed potential sources of variation across numerous studies. Through the analysis of 1299 observations garnered from 41 studies, we revealed the significant toxicity of PBDEs on plants. This result was verified to be robust and showed no signs of bias. Our study affirmed that functional indexes can contribute to variations that lead to the toxicity of PBDEs on various plants. Furthermore, we found that lower congeners PBDEs were more toxic to plants than higher congeners PBDEs, and higher plants were more resistant to PBDEs induced phytotoxicity than lower plants. For interactive effects, only specific PBDEs concentrations had significant effects on glutathione S-transferase activities, and experimental durations had no significant impacts on any functional indexes. These results reconciled empirical studies and assisted us with elucidating the ecotoxicology of PBDEs induced phytotoxicity.

© 2019 Elsevier Ltd. All rights reserved.

1. Introduction

As bromine flame retardants with excellent thermal stability, polybrominated diphenyl ethers (PBDEs) have been widely employed in plastic products, electronic appliances, and building materials on a global scale (Hu et al., 2007). Generally, PBDEs may be divided into higher PBDEs congeners (octa-BDE to deca-BDE) and lower PBDEs congeners (tri-BDE to hepta-BDE), based on their bromination numbers (Noyes et al., 2011). Due to the lack of the binding effects of its chemical bonds, PBDEs can be easily released into the environment (Alaee, 2003). Moreover, the toxic effects of PBDEs may freely proliferate through food chains (Hu et al., 2010), thereby threatening the security and stabilization of

Abbreviations: APx, ascorbate peroxidase; AsA, ascorbate; BDE-15, 4, 4' - dibromodiphenyl ether; BDE-28, 2, 4, 4' - tribrominated diphenyl ether, BDE-47; 2, 2', 4, 4' - tetrabrominated diphenyl ether, BDE-71; 2, 3', 4, 6' - tetrabrominated diphenyl ether, BDE-99; 2, 2', 4, 4', 5 - pentabromodiphenyl ether, BDE-153; 2, 2', 4, 4', 5, 5' - hexabrominated diphenyl ether; BDE-209, decabromodiphenyl ether; CAT, catalase; Chl, chlorophyll; DHA, dehydroascorbate; Fv/Fm, maximal efficiency of PSII photochemistry; GPx, glutathione peroxidase; GR, glutathione reductase; GSH, glutathione; GST, glutathione S-transferase; MDA, malondialdehyde; NR, nitrate reductase; PBDEs, Polybrominated diphenyl ethers; PI, perform index; POD, peroxidase; qP, photochemical quenching coefficient; ROS, reactive oxygen species; SD, standard deviation; SOD, superoxide dismutase.

^{*} Corresponding author.

E-mail address: hhruan@njfu.edu.cn (H. Ruan).

¹ These authors contributed equally to this work.

ecosystems (Yang et al., 2015; Zheng et al., 2015).

Many studies have reported that PBDEs are characterized by persistence, lipophilicity, and bioaccumulation, which seriously endangers humans and animal health (Maranghi et al., 2013; Shang et al., 2016). Although studies on the phytotoxicity of PBDEs have only begun in recent years (Sun et al., 2016), they were discovered to be toxic to most plants, which is manifested through the excessive production of reactive oxygen species (ROS) concentrations in plant cells (Wang et al., 2014; Xu et al., 2015). This leads to membrane lipid peroxidation with an increased content of malondialdehyde (MDA) (a lipid peroxidation product) (Xu et al., 2015), inactivation of enzymatic defense systems (Sun et al., 2019), and the destruction of plant cell structures with decreased protein content and damaged cell membranes (Qiu et al., 2018), which has adverse effects on nominal physiological and metabolic processes. Further, the decreased maximal efficiency of PSII photochemistry (Fv/Fm), performance index (PI), and chlorophyll (Chl) content, which reflected inhibited photosynthetic activities, were also found in plant leaves (Sun et al., 2016), which could eventually lead to inhibited plant growth, and even poisoning death (Kallqvist et al., 2006; Wang et al., 2014). However, several species of plants are resistant to PBDEs contamination to a significant degree. Since they can reduce the content of PBDEs in the environment through self-absorption or degradation (Huang et al., 2011; Wang et al., 2011b), they may be implemented for the phytoremediation of PBDEs contamination in the ambient environment (Mueller et al., 2006).

The challenge remains to comprehensively elucidate how higher and lower plants respond to PBDEs induced phytotoxicity, since the toxic effects of PBDEs on plants can vary substantially (Meng et al., 2018). For example, 14d-exposure of 5 µg/L 2,2',4,4'-tetrabromodiphenyl ether (BDE-47) demonstrated significant toxicity to *Lemna minor* (Qiu et al., 2018) and 56d-exposure of 5 mg/L BDE-47 caused significant toxicity to *Kandelia obovata* (Wang et al., 2014). Similarly, 5–10 mg/L of decabromodiphenyl ether (BDE-209) had a stimulating effect on *L. minor* after 14 days (Sun et al., 2019), whereas 100 mg/kg BDE-209 significantly inhibited the growth of ryegrass (*Lolium perenne* L) after 28 days (Xie et al., 2013). In a first attempt to quantify the phytotoxicity induced by PBDEs, it was discovered that BDE-47 could significantly inhibit the growth of *Skeletonema costatum* (Kallqvist et al., 2006).

Numerous studies have been published to illustrate the responses of plants to PBDEs induced phytotoxicity (Kallqvist et al., 2006; Meng et al., 2018; Qiu et al., 2018; Sun et al., 2019). These studies included different functional indexes (e.g., Fv/Fm, PI, Chl, superoxide dismutase (SOD), peroxidase (POD), catalase (CAT), and ROS), PBDEs concentrations, and experimental durations. PBDEs congeners and different plant organs were also considered in these experiments. Although a number of results are available, many of these appeared to be uncertain and controversial. Therefore, it was necessary to conduct a systematic analysis to summarize the responses of plants to PBDEs induced phytotoxicity. A meta-analysis is a statistical technique for generalizing the outcomes of multiple independent experiments that address the same subject. To the best of our knowledge, the present meta-analysis is the first to specifically focus on the toxicity of PBDEs on plants.

We hypothesized that, (i) PBDEs have significant toxicity for plants, (ii) the responses of plants to PBDEs induced phytotoxicity varies among functional indexes, vegetation types, plant organs, PBDEs congeners, experimental durations, and PBDEs concentrations, (iii) lower PBDEs congeners may initiate stronger toxicity to plants than higher PBDEs congeners, whereas higher plants are more tolerant to the phytotoxicity induced by PBDEs than lower plants. To test these hypotheses, we conducted a hierarchical meta-analysis that considered the nested structure of data, due to the

non-independence of multiple observations emanating from the same study (Rossetti et al., 2017).

2. Materials and methods

2.1. Data collection

A survey of online databases via Science Direct, Web of Science, and the China National Knowledge Infrastructure (CNKI) was conducted. The search included published articles from 1991 to January 2019. Different keyword combinations were used: PBDEs* OR BDE* AND toxicity* OR toxic effects* OR phytotoxicity* OR plants*. To be included in this research, the articles were required to be in both English, Chinese and Korean, and published in peer-reviewed journals and theses. We applied three criteria to determine whether a particular study should be included in the meta-analysis. First, the means and sample sizes (n) of the treatment groups and control groups must be reported. Second, only the data pertaining to PBDEs treatments and control groups were selected if the experiment included other treatments. Third, the PBDEs treatment durations and concentrations must be reported. A workflow diagram of the process and outcome of the study search is shown in Fig. 1.

In summary, for higher terrestrial plants (i.e., Chinese cabbage, maize, and *Oryza sativa*), the growth soil was artificially contaminated with a PBDEs stock solution and they were incubated in a greenhouse with an irradiance of 800–1400 µmol/m²/s, temperature of 23–25 °C, and light/dark cycle of 14 h/10 h. For higher aquatic plants (i.e., *L. minor*), the culture solution was Hoagland's solution (Hoagland and Arnon, 1950), whereas for lower aquatic plants (i.e., *Dicrateria inornata*, *Skeletonema costatum*, *Dunaliella salina*), the culture medium was F/2 medium (Guillard, 1975). The higher and lower aquatic plants were cultured with an irradiance of 80–100 µmol/m²/s, temperature of 22–26 °C, and light/dark cycle of 16 h/8h.

The experimental durations and PBDEs concentrations were converted to days and mg/L, respectively. When standard errors (SE) were presented in the paper, they were converted to SD by: $SD = SE \times \sqrt{n}$. The data presented in graphs were extracted by employing WebPlotDigitizer software (Burda et al., 2017). Following extraction, a total of 41 articles (representing 1299 observations) were identified (see Appendix S1; Dataset S1).

2.2. Moderator variables

Several explanatory variables affect the impacts of PBDEs on plants. These moderators were classified as categorical moderators and continuous moderators (Kinlock et al., 2018). Continuous moderators included experimental durations (days) and PBDEs concentrations (mg/L). Categorical moderators included functional indexes (i.e., growth, protein, Fv/Fm, Chl, MDA, ROS, glutathione S-transferase (GST), CAT, POD, and SOD); plant organs, classified as whole plant, leaf, stem, root, and cell; PBDEs congeners, classified as higher PBDEs congeners including BDE-209, and BDE-153; or lower PBDEs congeners, including BDE-99, BDE-71, BDE-47, BDE-28, and BDE-15; and vegetation types, segregated as higher terrestrial plants, higher aquatic plants, and lower aquatic plants.

2.3. Statistical analyses

Log response ratios (lnRR) were calculated as measures of the sizes of effects (Hedges et al., 1999):

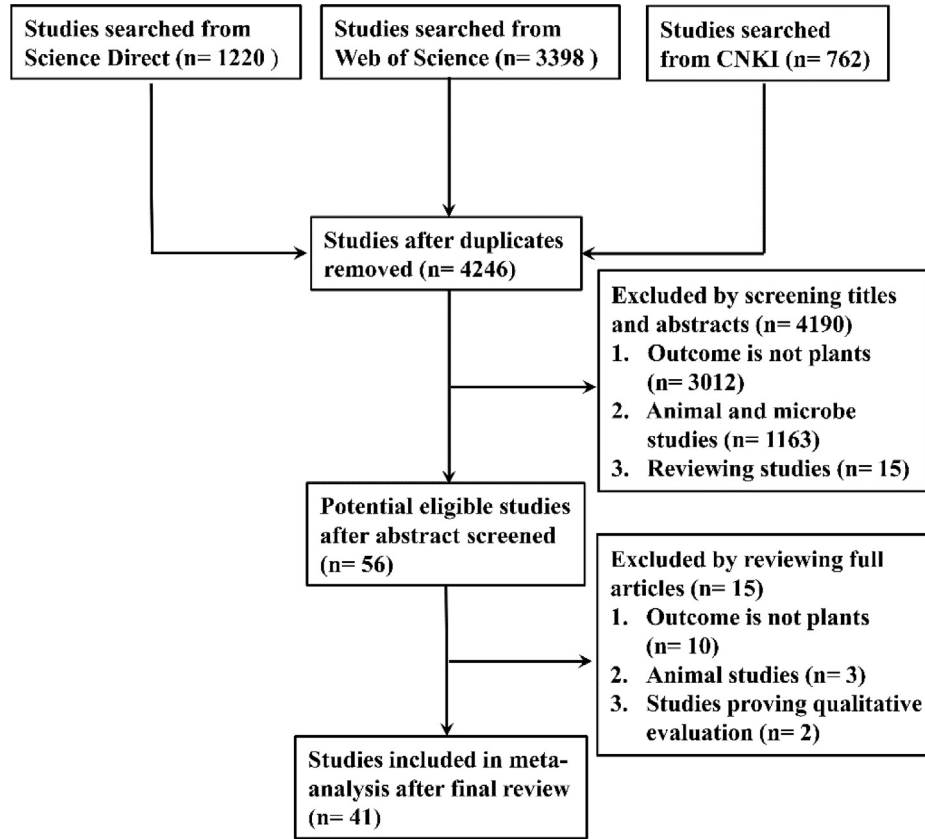


Fig. 1. Workflow diagram showing the procedure for selecting publications.

$$\ln RR = \ln \left(\frac{X_t}{X_c} \right) \quad (1)$$

where, X_t and X_c are the mean values for the PBDEs treatments and control groups, respectively. A value of $\ln RR \leq 0$ or $\ln RR > 0$ indicates whether the PBDEs treatment has a negative or positive effect on a plant. The variance (v) associated with each $\ln RR$ was calculated as:

$$V = \frac{S_t^2}{n_t X_t^2} + \frac{S_c^2}{n_c X_c^2} \quad (2)$$

where S_t and S_c represent the SD of the PBDEs treatments and control groups, respectively, with n_t and n_c as sample sizes. Not all cases reported SD or SE values, with 87 missing SD. In this situation, the missing SD could be attributed to the “Bracken, 1992” approach (Bracken, 1992; Benitez-Lopez et al., 2017). This analysis was performed through the application of *metagear* (Lajeunesse, 2016).

Further, we conducted a hierarchical mixed effects meta-analysis in *metafor* 2.0.0 (*rma.mv*) (Viechtbauer, 2010), which allowed for the explanation of nested groups. We specified reference and observation identities as random effects in our models (Peng et al., 2019).

The mean response ratio ($\ln RR_{++}$) was estimated from the $\ln RR$ of individual pairwise comparisons between the PBDEs treatments and control groups (Zhou et al., 2018):

$$\ln RR_{++} = \frac{\sum_{i=1}^m \sum_{j=1}^k w_{ij} \ln RR_{ij}}{\sum_{i=1}^m \sum_{j=1}^k w_{ij}} \quad (3)$$

where, $\ln RR_{++}$ is the j th observations ($j = 1, 2, \dots, k$) in the i th study

($i = 1, 2, \dots, m$), and w is the weight calculated by the reciprocal of variance of each $\ln RR$ ($1/v$) (Yue et al., 2017). Simultaneously, we obtained an estimate of the effect sizes for each study using a random-effects model with the *metafor* package (Demalach et al., 2017). The cumulative effect size represented the global scale of the effect present in each study. When 95% CI of $\ln RR_{++}$ did not overlap with zero, the treatment had significant effects on the variables (Koricheva et al., 2013). The percentages of changes in the variables initiated by the PBDEs treatments were calculated using the formula:

$$\text{Percentage}(\%) = [1 - \exp(\ln RR)] \times 100\% \quad (4)$$

Since a candidate set of moderators may contribute to variations in the impacts of PBDEs treatments on plants, it was necessary to compare all candidate models (Marino et al., 2018). Therefore, an information-theoretical approach was employed to select the best models to explain the differences in experimental outcomes (Burnham et al., 2011). The *glmulti* package was used for model selection and averaging, based on Akaike's information criterion corrected (AICc) (Calcagno and de Mazancourt, 2010). Once the full model was fitted (final number of models, 2^a , a : moderator numbers), we ranked the models based on their AICc values, and restricted further analyses to the models within $2 \Delta AICc$ from the models with the minimum AICc (Burnham et al., 2011).

We found that the effects of PBDEs on plants were related only to functional indexes (Table S1); thus, we selected functional indexes to discern the main effects via the omnibus test (Q_M). For each categorical variable, the total heterogeneity between all studies (Q_T) was segregated as between-group heterogeneity (Q_B) and within-group heterogeneity (Q_W). The Q_B was associated with P -values to model the heterogeneity of effect sizes that were related

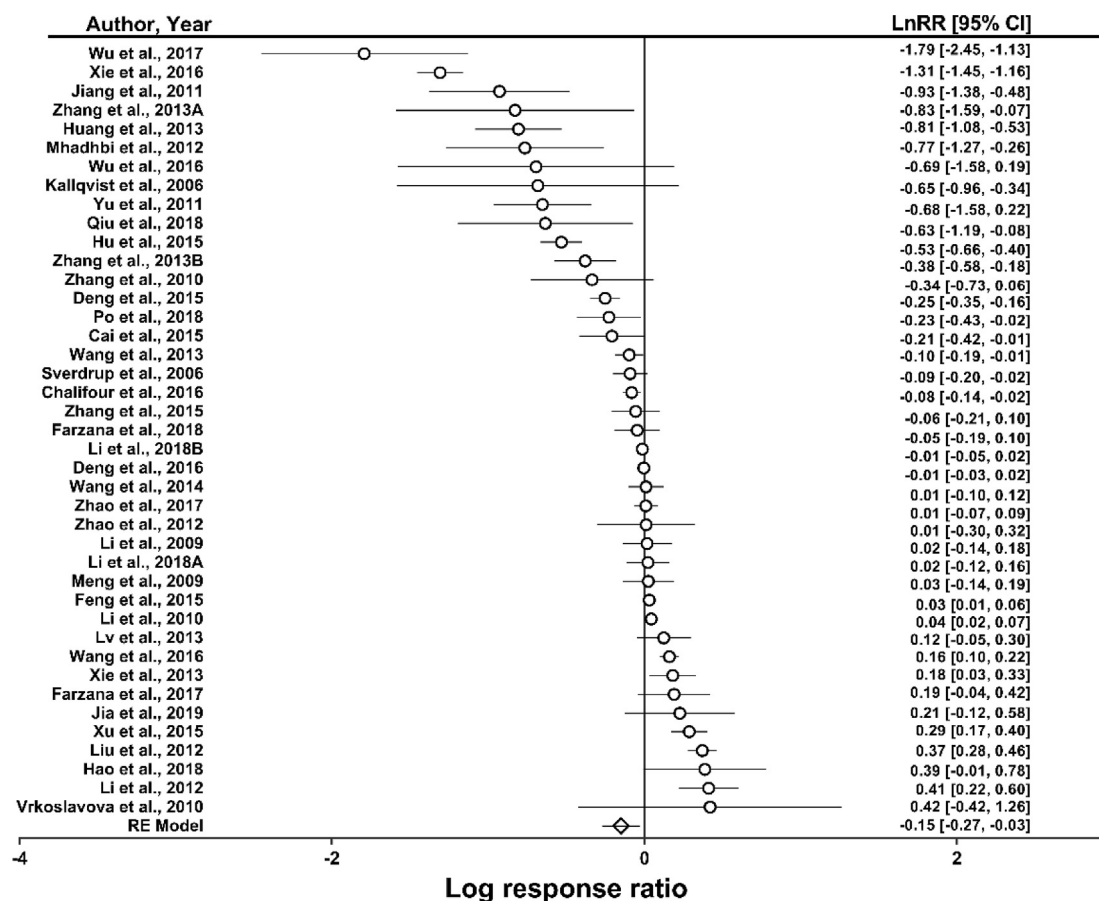


Fig. 2. Effect size estimates and 95% confidence intervals (CIs) for the effects of PBDEs on plants across different studies. The grand mean, represented by a rhombus, is at the bottom of the figure.

to variations between categories to obtain categorical comparisons. To further examine the core hypothesis, we tested the main effects of plant organs, PBDEs congeners, and vegetation types on phytotoxicity induced by PBDEs, and performed interactive effects with the functional index as the only fixed factor, and PBDEs concentrations and experimental durations as the interactive factors, respectively. We examined the relationships between effect sizes and continuous factors using a simple regression, to model one moderator at a time. All parameter estimates were analysed for the models and performed using the restricted maximum likelihood method with the *metafor* package. We ran all analyses with R 3.5.1 software.

2.4. Publication bias

The publication bias was tested graphically via a funnel plot in *rma.mv* function (Egger et al., 1997), the symmetry of which indicated its absence. We also calculated Rosenberg's fail-safe numbers to estimate the number of potential non-significant studies that were required to be added to impact our conclusion (Rosenberg, 2005). If the fail-safe number was $> 5k + 10$, where k represented the original number of observations, we could arrive at confident results (Rosenthal, 1991).

3. Results

3.1. Publication selection and database

The search string yielded 1220 articles via Science Direct, 3398

articles via Web of Science, and 762 articles from CNKI. Following the removal of duplicates, 4246 articles were selected for abstract screening. Of these, 4190 articles were excluded, as the outcome was not related to plants, or the articles were animal and microbe studies, or reviews. Subsequently, 15 articles were excluded after full text screening, leaving a total of 41 articles for our meta-analysis (Fig. 1).

Many studies exhibited more than one entry in the database (Dataset S1); thus, the total number of observations was far higher than the numbers of articles found. In this case, there were articles that involved more than one plant species, or observations for which the outcomes were separately presented by experimental durations, PBDEs concentrations, functional indexes, or plant organs. Notably, of all the observations, 426 experiments were performed with higher PBDEs congeners, and 873 experiments with lower PBDEs congeners. Regarding plants, 681 experiments employed higher terrestrial plants, 583 experiments used lower aquatic plants, whereas only 35 experiments used higher aquatic plants.

3.2. Evaluation of total heterogeneity

Across all collected studies, PBDEs exposure was found to cause an overall toxic effect on the physiological and biochemical functionalities of plants ($P = 0.02$; Fig. 2). The cumulative size of effects presented a negative value with a weighted LnRR_{++} of -0.15 [$\text{CI} = (-0.27, -0.03)$], which was also observed in most of the individual studies (Fig. 2). Moreover, we arrived at a reliable estimate regarding the effects of PBDEs on plants (funnel plot and

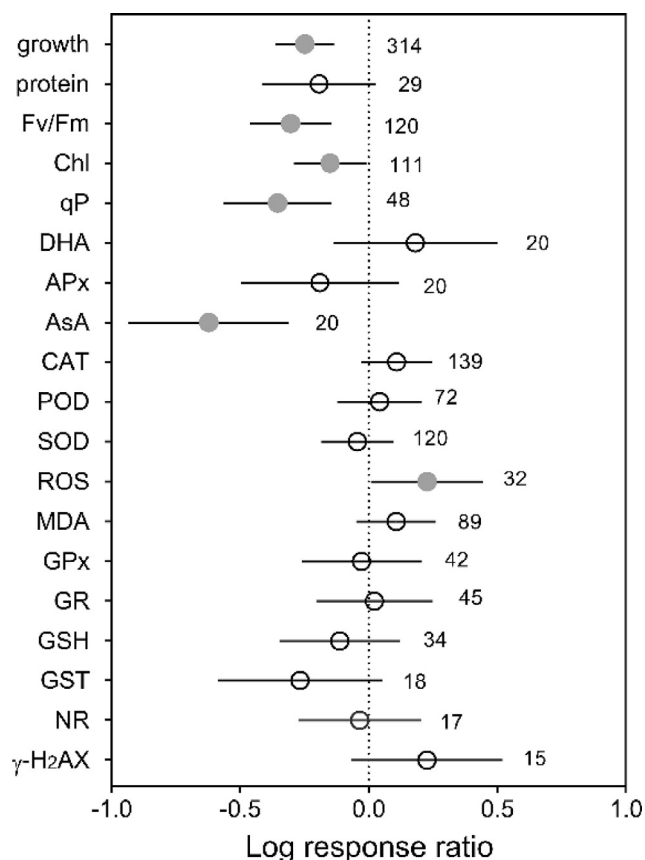


Fig. 3. Changes in toxicity of PBDEs on plants for different functional indexes. Dashed black line symbols refer to a log response ratio = 0. Error bar means 95% CIs. Grey and empty circles represent moderators with 95% CIs that do not overlap zero ($P < 0.05$), and overlap zero ($P > 0.05$), respectively. Numbers indicate sample size.

Rosenberg's fail-safe numbers, Appendix. S2).

3.3. Factors affecting plant responses to PBDEs-induced phytotoxicity

Functional indexes had a significant impact on the effects of PBDEs on plants ($Q_B = 185.21$, $P < 0.0001$; Table S2). In Fig. 3, we can see that PBDEs treatments significantly reduced the parameters of growth, Fv/Fm, Chl, photochemical quenching coefficient (qP), and ascorbate (AsA) by 22.09%, 26.29%, 13.98%, 29.89%, 46.33%, and 76.41%, with $\ln RR_{++}$ of -0.25 [CI = $(-0.37, -0.13)$], -0.31 [CI = $(-0.46, -0.15)$], -0.15 [CI = $(-0.29, -0.01)$], -0.36 [CI = $(-0.57, -0.14)$], and -0.62 [CI = $(-0.93, -0.31)$], respectively. Interestingly, only the ROS level showed a positive trend, increasing by 24.61%, with $\ln RR_{++}$ of 0.22 [CI = $(0.01, 0.44)$] under PBDEs exposure. Further, differences in the main effects of PBDEs on plants were not related to plant organs ($Q_B = 6.30$, $P = 0.18$; Table S2), PBDEs congeners ($Q_B = 5.49$, $P = 0.48$; Table S2), and vegetation type ($Q_B = 4.17$, $P = 0.12$; Table S2). However, cells, lower PBDEs congeners, and lower aquatic plants decreased by 22.77%, 15.63%, and 22.77%, with $\ln RR_{++}$ of -0.26 [CI = $(-0.42, -0.09)$], -0.17 [CI = $(-0.30, -0.04)$], and -0.26 [CI = $(-0.42, -0.09)$], respectively (Fig. 4; Table S2). For interactive effects, only the GST effect sizes increased significantly with elevated PBDEs concentrations ($P = 0.002$; Fig. 5A, Table S3), whereas other functional indexes did not differ significantly (Fig. 5A; Table S3). No functional indexes changed significantly with longer experimental duration (days) (Fig. 5B; Table S3).

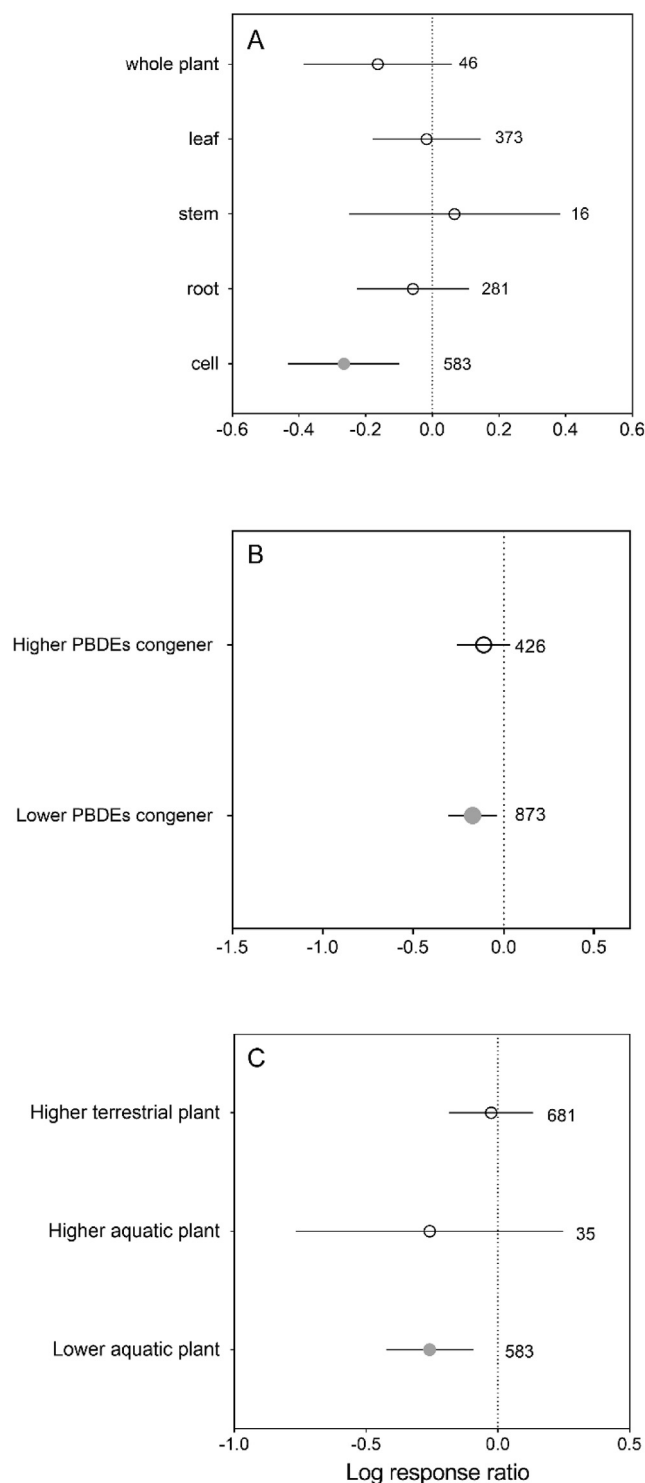


Fig. 4. Changes in toxicity of PBDEs on plants for different plant organs (A), PBDEs congeners (B), and vegetation types (C). Dashed black line symbols refer to a log response ratio = 0. Error bar means 95% CIs. Grey and empty circles represent moderators with 95% CIs that do not overlap zero ($P < 0.05$), and overlap zero ($P > 0.05$), respectively. Numbers indicate sample size.

4. Discussion

To the best of our knowledge, for the first time, the current meta-analysis based on 1299 observations, investigated the responses of plants to PBDEs treatments. In alignment with our

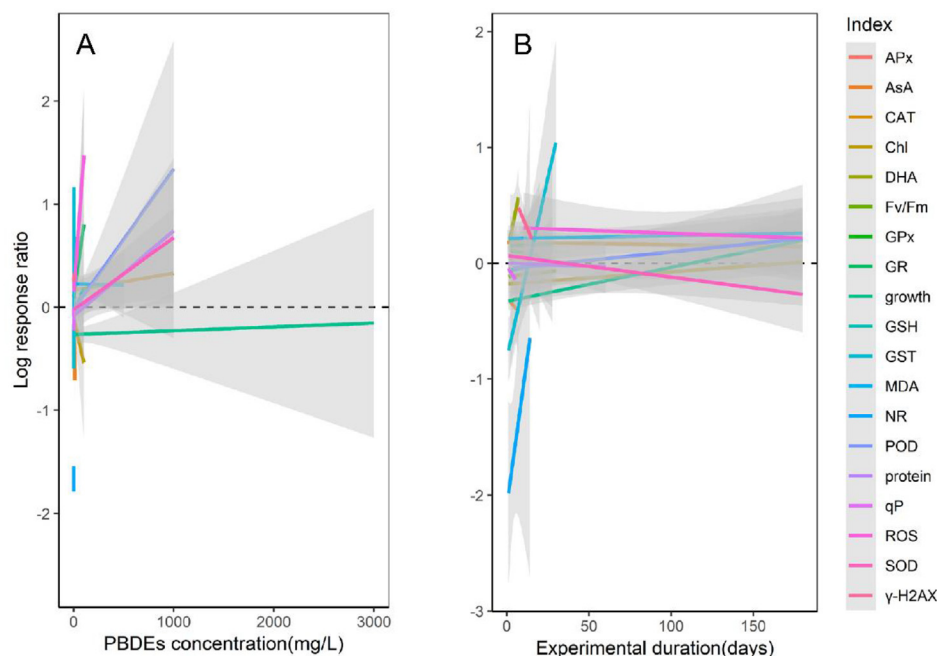


Fig. 5. Interactive effects between PBDEs concentrations (mg/L) and functional indexes (A); experimental duration (days) and functional indexes (B). Lines are fitted by linear-regression with 95% CIs in shade. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

hypothesis (i), we found that exposure to PBDEs had obvious toxicity to plants, which was, to some extent, consistent with Meng et al. (2018), who reported that PBDEs were toxic to most plants. According to our hypothesis (ii), we were able to identify that functional indexes contributed most to the toxicity of PBDEs. In addition, consistent with our hypothesis (iii), we found that lower PBDEs congeners initiated stronger phytotoxicity than the higher PBDEs congeners did, and lower plants were more sensitive to PBDEs induced phytotoxicity than higher plants. These results facilitated our understanding toward making more accurate predictions regarding the phytotoxicity of PBDEs.

4.1. Mechanisms associated with the effects of PBDEs on plants

Exposure to PBDEs impacted the physiological and biochemical functionality of plants differently, based on various functional indexes (Fig. 3). Variations in different functional indexes affected the growth and development of plants. It is well known that decreases in Fv/Fm, PI, Chl content, and qP characterize the suppression of photosynthetic functions (Zhao et al., 2017). However, increases in MDA and ROS content indicate that plants experienced oxidative stress, which may lead to the damage of cellular structures and physiological functions (Lv et al., 2015). Simultaneously, ROS-scavenging systems (e.g., SOD, POD, and CAT), and antioxidants (e.g., GST, dehydroascorbate (DHA), ascorbate peroxidase (APx), and AsA) are activated to clear excessive ROS (Huang et al., 2010b, 2013; Ke et al., 2011). Notably, the inhibited activities of ROS-scavenging systems and antioxidants revealed that the detoxification of PBDEs was limited (Qiu et al., 2018). Moreover, the increased γ-H2AX suggested DNA damage in plants (Xu et al., 2015).

In the present meta-analysis, we found significant decreases in the growth, Fv/Fm, Chl, and qP of plants under PBDEs treatments on a global scale (Fig. 3). These effects were not surprising given that growth and photosynthetic activity can be suppressed under PBDEs exposure, which were previously reported (Sun et al., 2016; Meng et al., 2018). The kinetics involved may include the capacity for PBDEs to directly damage thylakoid membranes (Qiu et al., 2018),

which can have further negative effects on photosynthetic functionality and growth. Further, PBDEs can directly impact the biofilm functions of chloroplasts and mitochondria (Pereira et al., 2014), leading to disorders in the electron transfer of chloroplasts and mitochondria, thus causing electron leakage and generating excessive ROS (Begović et al., 2016). Interestingly, ROS were the 'odd-one out', with an increased effect size under PBDE treatments, which was consistent with Sun et al. (2019) and Xie et al. (2013), who reported that PBDEs could lead to higher ROS concentrations in treated plants, in contrast to control groups. This 'odd-one out' pattern was also regarded as one of the primary PBDEs toxicity mechanisms (Wang et al., 2014).

To eliminate excessive ROS, antioxidants such as AsA, APx, and DHA were critical for plants to maintain metabolic stability (Huang et al., 2010a). Decreased AsA may have been attributed to its oxidation to DHA via excessive ROS generated by PBDEs (Wang and Tam, 2016). The decreased AsA and constant DHA, leading to a decreased AsA/DHA, suggested that the high consumption of AsA serves as an antioxidant to protect plants against oxidative stress (Rai et al., 2004). On the other hand, the responses of SOD, POD, CAT, glutathione (GPx), glutathione reductase (GR), glutathione (GSH), and nitrate reductase (NR) were not affected by PBDE treatments. This indicated that ROS scavenging systems and antioxidants were not inhibited by PBDEs, which suggested that plants possessed the basic capacity to scavenge the produced ROS (Torres et al., 2006); thus, the MDA content was not increased under PBDEs exposure. Additionally, NR and GST had been considered as participating in the degradation of PBDEs in plants (Qiu et al., 2018). We found a significant increase in GST and no changes in NR with incremental increases in PBDEs concentrations (Fig. 5A, Table S2), which suggested the role of GST in PBDEs degradation in plants (Huang et al., 2013).

Differences in the effects of cells, lower PBDEs congeners, and lower aquatic plants also explained some of the variations in the experimental results (Fig. 4). Together, these outcomes revealed that cells were the primary target of PBDEs in plants, where higher plants were more resistant to PBDEs induced phytotoxicity, and

higher PBDE congeners were less toxic to plants than lower PBDEs congeners, which was not unexpected. Huang et al. (2013) reported that the benzene rings of lower PBDEs possessed fewer bromine atoms and lower steric hindrance, which were more easily absorbed by cells. Interestingly, the experimental durations (days) had no significant impacts on the functional indexes under PBDEs induced phytotoxicity (Fig. 5B; Table S3). This was likely attributed to the adaptive strategies of plants to the phytotoxicity induced by PBDEs. First, in response to PBDEs exposure, the GST of plants will be activated with increased PBDEs concentrations to degrade them (Fig. 5A). In addition, Chl loss may be a step in an adaptive strategy to minimize the antenna size of photosystem II; thus, reducing the possibility of further injury being caused by excessive ROS (Kranter et al., 2002; Husen et al., 2014).

Various plants have different responses to the phytotoxicity induced by PBDEs. Some plants (e.g., *Pennisetum alopecuroides*, *Solanum nigrum*, *Ipomoea aquatica* Forsk. and *Chlorella*) have been observed to absorb PBDEs effectively; thus, they can be applied for the removal of PBDEs in the natural environment (Liu et al., 2012; Deng and Tam, 2016). However, many plants (e.g., *L. minor*, *Skeletonema costatum*, *Heterosigma akashiwo*, and *Karenia mikimotoi*) are sensitive to the phytotoxicity induced by PBDEs and can be used as model plants for phytotoxicity research under laboratory conditions, such that they may be selected as ideal indicator plants for PBDEs pollution in the ambient environment (Kallqvist et al., 2006; Zhang et al., 2013; Sun et al., 2019). Currently, the phytotoxicity induced by PBDEs in natural ecosystems is rarely investigated, therefore, field experiments involving PBDEs induced phytotoxicity require further in-depth study.

4.2. Implications for future experiments regarding the effects of PBDEs on plants

The results of this meta-analysis revealed that our current knowledge of the responses of plants to PBDEs induced phytotoxicity is far from satisfactory. In the natural environment following long-term PBDEs exposure, the order of magnitude of total PBDEs concentrations was found to be only in the ng/L (Yang et al., 2015) or µg/kg FW range (Wang et al., 2011a). However, the experimental durations and PBDEs concentrations of the studies included in our meta-analysis concentrated on a 10–40 days duration and 0.001–7 mg/L concentrations (Dataset S1). Consequently, we propose that longer-term (>50 days) experiments and lower PBDE concentrations should be established, in that brief experimental durations and high PBDEs concentrations may confuse the assessment of PBDEs induced phytotoxicity characterized by persistence and bioaccumulation (Shang et al., 2016). Further, different congeners of PBDEs are frequently detected in the environment (Law et al., 2006); therefore, future experiments should focus more on the combined effects of different PBDEs congeners on plants.

5. Conclusions

This meta-analysis demonstrated that PBDEs treatments imparted significant toxicity to plants. The phytotoxicity induced by PBDEs was varied and contingent on functional indexes. Plant cells, lower PBDEs congeners, and lower aquatic plants were also key moderators that determined the responses of plants to PBDEs. GST activity can be enhanced with elevated PBDEs concentrations (mg/L). Further, the experimental durations (days) had no significant influences on any functional indexes under exposure of PBDEs. Longer-term experiments combined with the quantification of the phytotoxicities of different PBDEs congeners under low concentrations are urgently required toward better elucidating and more accurately predicting how plants respond to PBDEs induced

phytotoxicity.

Acknowledgements

This work was supported by the National Key Research and Development Program of China (No. 2016YFD0600204) and Priority Academic Program Development of Jiangsu Higher Education Institutions (PAPD).

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chemosphere.2019.124865>.

References

- Alaee, M., 2003. An overview of commercially used brominated flame retardants, their applications, their use patterns in different countries/regions and possible modes of release. *Environ. Int.* 29, 683–689.
- Begović, L., Mlinarić, S., Antunović Dunić, J., Katanić, Z., Lončarić, Z., Lepeduš, H., Cesar, V., 2016. Response of *Lemna minor* L. to short-term cobalt exposure: the effect on photosynthetic electron transport chain and induction of oxidative damage. *Aquat. Toxicol.* 175, 117–126.
- Benitez-Lopez, A., Alkemade, R., Schipper, A.M., Ingram, D.J., Verweij, P.A., Eikelboom, J.A.J., Huijbregts, M.A.J., 2017. The impact of hunting on tropical mammal and bird populations. *Science* 356, 180–183.
- Bracken, M.B., 1992. *Effective Care of the Newborn Infant*. Oxford Univ. Press, Oxford.
- Burda, B.U., O'Connor, E.A., Webber, E.M., Redmond, N., Perdue, L.A., 2017. Estimating data from figures with a Web-based program: considerations for a systematic review. *Res. Synth. Methods* 8, 258–262.
- Burnham, K.P., Anderson, D.R., Huyvaert, K.P., 2011. AIC model selection and multimodel inference in behavioral ecology: some background, observations, and comparisons. *Behav. Ecol. Sociobiol.* 65, 23–35.
- Calcagno, V., de Mazancourt, C., 2010. Glumtl: an R package for easy automated model selection with (generalized) linear models. *J. Stat. Softw.* 34, 1–29.
- Demalch, N., Zaady, E., Kadmon, R., 2017. Contrasting effects of water and nutrient additions on grassland communities: a global meta-analysis. *Glob. Ecol. Biogeogr.* 26, 1–10.
- Deng, D., Tam, N.F.Y., 2016. Adsorption-uptake-metabolism kinetic model on the removal of BDE-47 by a *Chlorella* isolate. *Environ. Pollut.* 212, 290–298.
- Egger, M., Davey Smith, G., Schneider, M., Minder, C., 1997. Bias in meta-analysis detected by a simple, graphical test. *BMJ Br. Med. J. (Clin. Res. Ed.)* 315, 629–634.
- Guillard, R.R.L., 1975. *Culture of Phytoplankton for Feeding Marine Invertebrates*. Plenum Press, New York.
- Hedges, L.V., Gurevitch, J., Curtis, P.S., 1999. The meta-analysis of response ratios in experimental ecology. *Ecology* 80, 1150–1156.
- Hoagland, D.R., Arnon, D.I., 1950. The water-culture method for growing plants without soil. *Calif. Agric. Ext. Serv. Circ.* 347, 30–31.
- Hu, C.C., Dai, J.Y., Xu, Z.C., Luo, X.J., Cao, H., Wang, J.S., Mai, B.X., Xu, M.Q., 2010. Bioaccumulation behavior of polybrominated diphenyl ethers (PBDEs) in the freshwater food chain of Baiyangdian lake, north China. *Environ. Int.* 36, 309–315.
- Hu, X.Z., Xu, Y., Hu, D.C., Hui, Y., Yang, F.X., 2007. Apoptosis induction on human hepatoma cells Hep G2 of decabrominated diphenyl ether (PBDE-209). *Toxicol. Lett.* 171, 19–28.
- Huang, G.Y., Wang, Y.S., Sun, C.C., Dong, J.D., Sun, Z.X., 2010a. The effect of multiple heavy metals on ascorbate, glutathione and related enzymes in two mangrove plant seedlings (*Kandelia candel* and *Bruguiera gymnorhiza*). *Oceanol. Hydrobiol. Stud.* 39, 11–25.
- Huang, H., Zhang, S., Christie, P., Wang, S., Xie, M., 2010b. Behavior of decabromodiphenyl ether (BDE-209) in the Soil-Plant system: uptake, translocation, and metabolism in plants and dissipation in soil. *Environ. Sci. Technol.* 44, 663.
- Huang, H.L., Zhang, S.Z., Christie, P., 2011. Plant uptake and dissipation of PBDEs in the soils of electronic waste recycling sites. *Environ. Pollut.* 159, 238–243.
- Huang, H.L., Zhang, S.Z., Wang, S., Lv, J.T., 2013. In vitro biotransformation of PBDEs by root crude enzyme extracts: potential role of nitrate reductase (NaR) and glutathione S-transferase (GST) in their debromination. *Chemosphere* 90, 1885–1892.
- Husen, A., Iqbal, M., Aref, I.M., 2014. Growth, water status, and leaf characteristics of *Brassica carinata* under drought and rehydration conditions. *Braz. J. Bot.* 37, 217–227.
- Kallqvist, T., Grung, M., Tollefsen, K.E., 2006. Chronic toxicity of 2,4,2',4'-tetrabromodiphenyl ether on the marine alga *Skeletonema costatum* and the crustacean *Daphnia magna*. *Environ. Toxicol. Chem.* 25, 1657–1662.
- Ke, L., Zhang, C., Wong, Y.S., Tam, N.F., 2011. Dose and accumulative effects of spent lubricating oil on four common mangrove plants in South China. *Ecotoxicol. Environ. Saf.* 74, 55–66.
- Kinlock, N.L., Prowant, L., Herstoff, E.M., Foley, C.M., Akin-Fajiyi, M., Bender, N.,

- Umarani, M., Ryu, H.Y., Şen, B., Gurevitch, J., 2018. Explaining global variation in the latitudinal diversity gradient: meta-analysis confirms known patterns and uncovers new ones. *Glob. Ecol. Biogeogr.* 27, 125–141.
- Koricheva, J., Gurevitch, J., Mengersen, K., 2013. *Handbook of Meta-Analysis in Ecology and Evolution*. Princeton University Press.
- Kranner, I., Beckett, R.P., Wornik, S., Zorn, M., Pfeifhofer, H.W., 2002. Revival of a resurrection plant correlates with its antioxidant status. *Plant J.* 31, 13–24.
- Lajeunesse, M.J., 2016. Facilitating systematic reviews, data extraction and meta-analysis with the metagear package for R. *Methods Ecol. Evol.* 7, 323–330.
- Law, R.J., Allchin, C.R., de Boer, J., Covaci, A., Herzke, D., Lepom, P., Morris, S., Tronczynski, J., de Wit, C.A., 2006. Levels and trends of brominated flame retardants in the European environment. *Chemosphere* 64, 187–208.
- Liu, J., Yin, H., Peng, H., Ye, J., Ye, Q., Li, L., He, B., 2012. Research on the physiological responses of six plants *Pennisetum alopecuroides* etc to BDE-209 in soil and their phytoremediation effect. *J. Agro-Environ. Sci.* 31, 1745–1751.
- Lv, Q.Y., Wan, B., Guo, L.H., Zhao, L., Yang, Y., 2015. In vitro immune toxicity of polybrominated diphenyl ethers on murine peritoneal macrophages: apoptosis and immune cell dysfunction. *Chemosphere* 120, 621–630.
- Maranghi, F., Tassinari, R., Moracci, G., Altieri, I., Rasinger, J.D., Carroll, T.S., Hogstrand, C., Lundebye, A.K., Mantovani, A., 2013. Dietary exposure of juvenile female mice to polyhalogenated seafood contaminants (HBCD, BDE-47, PCB-153, TCDD): comparative assessment of effects in potential target tissues. *Food Chem. Toxicol.* 56, 443–449.
- Marino, N., Romero, G.Q., Farjalla, V.F., 2018. Geographical and experimental control modulate the effect of warming on top-down control: a meta-analysis. *Ecol. Lett.* 21, 455–466.
- Meng, Y., Zhang, W., Wang, J., Jiang, D., Qiu, N., 2018. Research progress of plant toxicology of polybrominated diphenyl ethers (PBDEs). *Plant Physiol. J.* 54, 183–191.
- Mueller, K.E., Mueller-Spitz, S.R., Henry, H.F., Vonderheide, A.P., Soman, R.S., Kinkle, B.K., Shann, J.R., 2006. Fate of pentabrominated diphenyl ethers in soil: abiotic sorption, plant uptake, and the impact of interspecific plant interactions. *Environ. Sci. Technol.* 40, 6662–6667.
- Noyes, P.D., Hinton, D.E., Stapleton, H.M., 2011. Accumulation and debromination of decabromodiphenyl ether (BDE-209) in juvenile fathead minnows (*Pimephales promelas*) induces thyroid disruption and liver alterations. *Toxicol. Sci.: Off. J. Soc. Toxicol.* 122, 265–274.
- Peng, S., Kinlock, N.L., Gurevitch, J., Peng, S.L., 2019. Correlation of native and exotic species richness: a global meta-analysis finds no invasion paradox across scales. *Ecology* 100.
- Pereira, L.C., Miranda, L.F., de Souza, A.O., Dorta, D.J., 2014. BDE-154 induces mitochondrial permeability transition and impairs mitochondrial bioenergetics. *J. Toxicol. Environ. Health Part A* 77, 24–36.
- Qiu, N., Wang, R., Sun, Y., Wang, X., Jiang, D., Meng, Y., Zhou, F., 2018. Toxic effects and mechanism of 2,2',4,4'-tetrabromodiphenyl ether (BDE-47) on *Lemna minor*. *Chemosphere* 193, 711–719.
- Rai, V., Vajpayee, P., Singh, S.N., Mehrotra, S., 2004. Effect of chromium accumulation on photosynthetic pigments, oxidative stress defense system, nitrate reduction, proline level and eugenol content of *Ocimum tenuiflorum* L. *Plant Sci.* 167, 1159–1169.
- Rosenberg, M.S., 2005. The file-drawer problem revisited: a general weighted method for calculating fail-safe numbers in meta-analysis. *Evolution* 59, 464–468.
- Rosenthal, R., 1991. *Meta-analytic Procedures for Social Research*. Sage, Newbury Park, California.
- Rossetti, M.R., Tschardtke, T., Aguilar, R., Batary, P., 2017. Responses of insect herbivores and herbivory to habitat fragmentation: a hierarchical meta-analysis. *Ecol. Lett.* 20, 264–272.
- Shang, X., Dong, G., Zhang, H., Zhang, L., Yu, X., Li, J., Wang, X., Yue, B., Zhao, Y., Wu, Y., 2016. Polybrominated diphenyl ethers (PBDEs) and indicator polychlorinated biphenyls (PCBs) in various marine fish from Zhoushan fishery, China. *Food Control* 67, 240–246.
- Sun, Y., Qiu, N., Wang, R., 2016. Toxic effect of 2,2',4,4'-tetrabromodiphenyl ether (BDE-47) on *Lemna minor*. *Plant Physiology Journal* 1576–1582.
- Sun, Y., Sun, P., Wang, C., Liao, J., Ni, J., Zhang, T., Wang, R., Ruan, H., 2019. Growth, physiological function, and antioxidant defense system responses of *Lemna minor* L. to decabromodiphenyl ether (BDE-209) induced phytotoxicity. *Plant Physiol. Biochem.: PPB (Plant Physiol. Biochem.)* 139, 113–120.
- Torres, M.A., Jones, J.D.G., Dangel, J.L., 2006. Reactive oxygen species signaling in response to pathogens. *Plant Physiol.* 141, 373–378.
- Viechtbauer, W., 2010. Conducting meta-analyses in R with the metafor package. *J. Stat. Softw.* 36, 1–48.
- Wang, S., Zhang, S.Z., Huang, H.L., Christie, P., 2011a. Behavior of decabromodiphenyl ether (BDE-209) in soil: effects of rhizosphere and mycorrhizal colonization of ryegrass roots. *Environ. Pollut.* 159, 749–753.
- Wang, S., Zhang, S.Z., Huang, H.L., Zhao, M.M., Lv, J.T., 2011b. Uptake, translocation and metabolism of polybrominated diphenyl ethers (PBDEs) and polychlorinated biphenyls (PCBs) in maize (*Zea mays* L.). *Chemosphere* 85, 379–385.
- Wang, Y., Tam, N.F.Y., 2016. Glutathione-ascorbate cycle is an early warning indicator of toxicity of BDE-47 in mangroves. *J. Environ. Qual.* 47, 212–220.
- Wang, Y., Zhu, H., Tam, N.F., 2014. Effect of a polybrominated diphenyl ether congener (BDE-47) on growth and antioxidative enzymes of two mangrove plant species, *Kandelia obovata* and *Avicennia marina*, in South China. *Mar. Pollut. Bull.* 85, 376–384.
- Xie, X., Qian, Y., Xue, Y., He, H., Wei, D., 2013. Plant uptake and phytotoxicity of decabromodiphenyl ether (BDE-209) in ryegrass (*Lolium perenne* L.). *Environ. Sci. Processes Impacts* 15, 1904–1912.
- Xu, X., Huang, H., Wen, B., Wang, S., Zhang, S., 2015. Phytotoxicity of brominated diphenyl ether-47 (BDE-47) and its hydroxylated and methoxylated analogues (6-OH-BDE-47 and 6-MeO-BDE-47) to maize (*Zea mays* L.). *Chem. Res. Toxicol.* 28, 510–517.
- Yang, Y., Xie, Q., Liu, X., Wang, J., 2015. Occurrence, distribution and risk assessment of polychlorinated biphenyls and polybrominated diphenyl ethers in nine water sources. *Ecotoxicol. Environ. Saf.* 115, 55–61.
- Yue, K., Fornara, D.A., Yang, W., Peng, Y., Li, Z., Wu, F., Peng, C., 2017. Effects of three global change drivers on terrestrial C:N:P stoichiometry: a global synthesis. *Glob. Chang. Biol.* 23, 2450–2463.
- Zhang, X., Tang, X., Zhou, B., Wang, Y., 2013. Effects of decabromodiphenyl ether (BDE-209) on inter-specific competition between two species of marine bloom-forming microalgae. *PLoS One* 8, 1–11.
- Zhao, Y., Wang, Y., Li, Y., Santschi, P.H., Quigg, A., 2017. Response of photosynthesis and the antioxidant defense system of two microalgal species (*Alexandrium minutum* and *Dunaliella salina*) to the toxicity of BDE-47. *Mar. Pollut. Bull.* 124, 459–469.
- Zheng, Q., Nizzetto, L., Li, J., Mulder, M.D., Sanka, O., Lammel, G., Bing, H., Liu, X., Jiang, Y., Luo, C., Zhang, G., 2015. Spatial distribution of old and emerging flame retardants in Chinese forest soils: sources, trends and processes. *Environ. Sci. Technol.* 49, 2904–2911.
- Zhou, Z.H., Wang, C.K., Luo, Y.Q., 2018. Effects of forest degradation on microbial communities and soil carbon cycling: a global meta-analysis. *Glob. Ecol. Biogeogr.* 27, 110–124.